Technoverse Hackathon

2026

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DriftVeil - AI Driven Machinery Drift Detection System | Business Plan (1/2)

WHY

Explain the Problem

Problem Description & Business Scenario : Manufacturing downtime costs **$50B annually**, and factories lose about **$125,000 per hour** when machines fail. Most plants use threshold-based systems that alert only after damage starts.
Small daily changes (e.g., a motor rising **0.3°C/day**) go unnoticed until sudden failure, this is the **silent drift problem**.
Problem Scope: Slow, hidden performance decline in machines like motors, pumps, conveyors, and bottling lines.
Multiple sensors degrade slightly together, but no single alarm triggers.
Target: While tailored for mid-size SME manufacturing lines, it natively scales to enterprise-level.
Target Users/Stakeholders: Plant Operators, Maintenance Engineers, Factory Owners, Production Managers

How

Explain the Solve

Solution Overview: DriftVeil is an AI platform that continuously calculates **telemetry drift** to identify hidden machine degradation before sudden failure. It utilizes **What-If simulation**s to test corrective actions predicting exactly how specific machine adjustments will impact time-to-failure, long before SCADA threshold alarms ever trigger.
Technical Details: The MVP operates natively within an **AWS environment**, streaming TimescaleDB telemetry via secure, edge-encrypted OT network gateways. **CUSUM** algorithms and **z-scores** track the drift slope with executing live cross-machine correlation. Four specialized **AWS Strands Agents** (powered by Amazon Bedrock) autonomously process these metrics via a secure **MCP-native architecture**, grounded by a continuous **fingerprint library feedback loop**. After aggregating 6 months (~1TB) of operational data per facility, our continuous feedback loop will safely upgrade to Deep Learning LSTMs.
Innovation: Crucially, we augment, rather than replace, incumbent SCADA infrastructure from legacy vendors like **Siemens and Rockwell.** Driven by design thinking, our differentiator eliminates black-box AI by catching the continuous slope rather than just the traditional SCADA anomaly spike. Pipeline: `[Live Telemetry]` → `[CUSUM Score]` → `[Bedrock Root-Cause Reasoning]` → `[Predictive ETA]`. **Bedrock Guardrails** guarantee zero hallucinations.
Market Potential: Validated by **McKinsey's $8B** Predictive Maintenance **TAM**, our scalable **B2B SaaS** architecture targets critical industry domains. Capturing an initial **2% SOM** drives high recurring revenue. By preventing an $80K manufacturing stoppage with a $2K scheduled repair, clients immediately realize our validated **4,000% ROI** for every early drift event caught.

WHAT

Value proposition

Primary benefits: Early drift detection; prevention of sudden breakdowns; proactive maintenance; improved equipment reliability; explainable insights that improve decision support, reduced disruption to operations; higher uptime
Efficiency and flexibility: Real-time monitoring; continuous sensor-based analysis; adaptable across machine types; flexible deployment across plants; easy integration with industrial systems; no replacement of existing hardware
Time and cost saving: Reduced downtime; lower repair costs; fewer emergency interventions; optimized maintenance scheduling; reduced manual inspection effort; lower production loss; extended machine lifespan
Scalability: Multi-machine deployment; line-level scalability; plant-wide adoption; modular architecture; enterprise-ready monitoring framework; future expansion across industries
Social Impact: Safer industrial operations; reduced resource wastage; improved sustainability; support for smart manufacturing

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Financials & Timelines | Business Plan (2/2)

Investments

What does it take & How much does it cost to solve?

Cloud: Amazon EC2, RDS, ElastiCache, Kinesis (~$2K–$5K/mo per plant)
AI Agents: AWS Strands, Amazon Bedrock, Amazon Nova, Anthropic Claude API (Pay-per-token API usage)
No hardware procurement, no SCADA replacement
Process Re-engg: Predictive maintenance (Zero CapEx), Automated Troubleshooting Workflows
Tools: MCP architecture, Fingerprint libraries, Next.js UI (Open-source / Included)
Frameworks: Pandas, NumPy, Scikit-learn, TimescaleDB, Redis, Kafka, Docker, PostgresSQL.

Returns

Quantify the benefits & What if I don't solve?

ROI: Reduced downtime; lower repair costs; better asset utilization
Revenue Growth: Higher uptime; smoother production; improved delivery reliability
Happy Customer: Fewer delays; more reliable output; better service consistency
Margin Improvement: Lower maintenance costs; reduced waste; improved efficiency
Employee Experience: Explainable alerts; less manual effort; faster decisions
Returns: 30–50% less downtime; **15–25%** maintenance cost savings; higher uptime and efficiency.
If not solved: hidden drift; sudden failures; rising costs; lower productivity

Timelines

Time to realize benefits

Weeks 1–2: Sensor integration, telemetry setup, dashboard live
Weeks 3–4: Baselines established, model training, first alerts with root-cause insights
Month 2: Multi-agent pipeline activated, predictive monitoring starts
Month 3: Pilot success, first intervention, savings documented
Month 4–5: Self-improving fingerprint library, expansion to more machines
Month 6: Scale-up across production lines, measurable ROI and reliability gains

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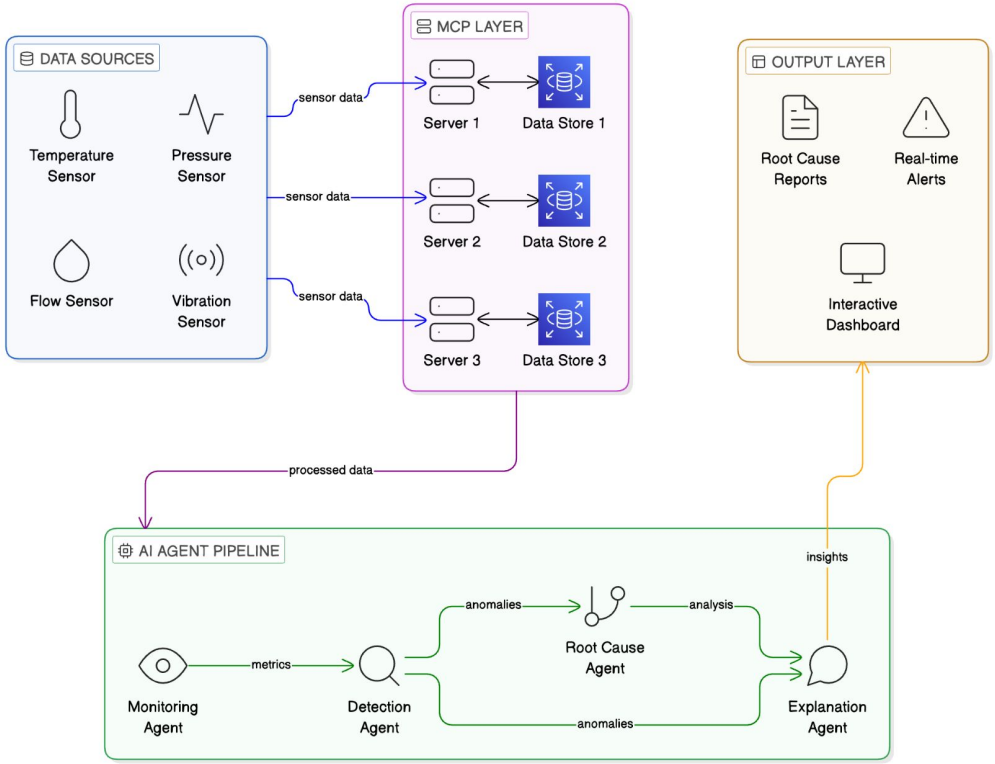
DriftVeil - AI Driven Machinery Drift Detection System | **Architecture Diagram**



- Sensor data flow
- Processed data to AI
- Agent pipeline flow
- Output delivery
- Data store sync

AI & Agent Layer

- AWS Strands —
- Amazon Bedrock —
- Anthropic Claude —
- Amazon Nova —
- MCP Model Context Protocol
- Bedrock Guardrails

Detection & Data

- scikit-learn
- NumPy
- Pandas
- TimescaleDB
- Redis
- SQLite

Frontend & Infrastructure

- Next.js
- Python
- Docker
- PostgreSQL
- Recharts
- shadcn/ui

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Thank you

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